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AI and Water:

Toward a Sustainable Digital Future

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As artificial intelligence (AI) becomes increasingly central to environmental services, its own environmental footprint – particularly water use – remains poorly understood. This paper examines the water footprint of AI through the lens of data centers, considering three key dimensions: direct water consumption from cooling systems, indirect water use embedded in the electricity that powers these facilities, and the indirect water embodied in hardware production. Training and deploying large-scale AI models demand immense computational power, which drives high energy use and substantial heat generation. To dissipate this heat, many data center facilities rely on water-intensive cooling systems that draw heavily on freshwater sources. While data centers' global water use is modest compared to major sectors such as agriculture, manufacturing, and energy, concerns are mounting about their rapid expansion and concentrated local impacts. Hyperscale facilities, in particular, can intensify localized water stress, especially in arid or water-scarce regions, creating competition with residential, industrial, and agricultural users and placing additional pressure on already fragile water systems. These risks highlight the urgency of aligning AI growth and data center development with local and regional water resource planning, environmental policy, and infrastructure resilience. This paper identifies technological and system-level barriers that hinder sustainable AI deployment and proposes strategies to align AI development and deployment with sustainable, water-conscious growth. Key priorities include fostering technological innovation and optimization (e.g., advancing cooling technologies, improving efficiency through model compression, chip design, algorithmic improvement, and operational and network optimization), strengthening water resource management practices (e.g., applying circular water economy principles such as wastewater reuse, water recovery, and closed-loop cooling, while better integrating data center planning into local and regional water management), and enhancing transparency through standardized water-use reporting at both the infrastructure and AI model levels. Ultimately, the paper calls for a multidisciplinary, systems-level approach to AI governance that aligns technological advancement with sustainable water stewardship, ensuring innovation supports – rather than undermines – long-term water security and resilience.

Introduction

AI is increasingly vital to environmental services. AI-powered IoT (Internet of Things) sensors help utilities detect leaks and abnormal usage in real time,[1] reducing water loss and improving demand forecasting by integrating weather, consumption, and socioeconomic data.[2] AI enhances water treatment efficiency by optimizing chemical and energy use while maintaining quality.[3] In agriculture, AI enables precision irrigation, significantly reducing water use without lowering yields.[4] Additionally, AI strengthens flood prediction by analyzing satellite imagery, weather models, and hydrological data in real time.[5]



Cooling tower

However, as AI systems continue to proliferate, their environmental impact is drawing growing concern. While energy use has traditionally dominated environmental discussions around AI, its water footprint is emerging as a critical, and often underrecognized, dimension of sustainability. As climate change makes water supplies increasingly unpredictable, managing water systems is becoming more challenging, intensifying the need to understand and reduce AI's local water footprint to support long-term resilience of both digital and natural systems.

This paper examines the water implications of AI operations across their lifecycle and at multiple scales. It introduces a framework for integrating water-conscious planning into the development and deployment of these systems. The analysis emphasizes the critical role of cooling technologies, energy sources, data transparency, and water management strategies, while advocating for sustainable innovation in AI design and operation. Ultimately, the paper outlines a pathway for measuring, managing, and reducing the water footprint of AI and data centers.

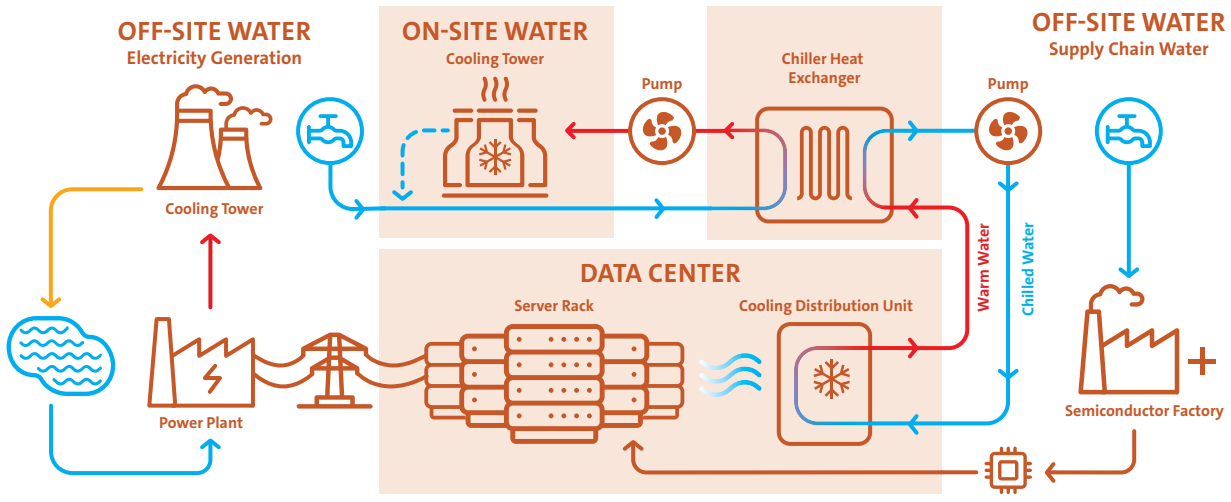
The Water Footprint of Data Centers

AI processes depend on high-performance servers housed in data centers, which consume large amounts of electricity and generate substantial amounts of heat. To prevent server overheating, many facilities use water-intensive cooling systems that rely on freshwater for heat dissipation. Although currently AI represents a small (~15%) portion of data center electricity use,[6] this technology is the fastest expanding workload in data centers[7] and is expected to account for 27% of their energy use by 2027.[8] To assess AI's water footprint, we use data centers' overall water use as a proxy, given the limited availability of AI-specific data and the intrinsic connection between AI workloads and data center operations.[9]

When tracking water footprints is it important to distinguish between water use, water withdrawals, and water consumption. In this article, we use water use as an umbrella term for all ways water is engaged in an activity or system, but we differentiate two specific quantities: *water withdrawals,* the volumes taken from a source (river, lake, or aquifer) to meet a need – some of which return as effluent or cooling blowdown – and *water consumption* (depletion), the share of withdrawn water effectively removed from the local basin in a usable time, place, or quality (e.g., through evaporation, incorporation into products, or discharge to another watershed). This distinction matters: infrastructure sizing and permits are driven by withdrawals, whereas ecological availability and drought resilience hinge on consumption, because consumed water is not readily available to other users or ecosystems.

In addition, data center water use occurs across multiple dimensions including direct and indirect water consumption (Figure 1):

Figure 1: *Direct and indirect water use in data centers*



Source: Adapted from Pengfei Li, Jianyi Yang, Mohammad A. Islam, and Shaolei Ren (2025).

- **Indirect Water Use Through Energy Generation:** Beyond on-site cooling, data centers also drive substantial indirect water use through electricity generation, primarily from thermoelectric and hydroelectric sources. According to the U.S. Geological Survey, thermoelectric power accounts for about 40% of national freshwater withdrawals.[12] A study by the International Energy Agency (EIA)[6] found that electricity generation is often the largest contributor to a data center's water footprint (Figure 2), with around two-thirds of total water use tied to power supply and just one-quarter to direct cooling.

  Choices in cooling technologies and energy sources have major water and energy implications, especially in water-stressed regions. These choices often involve trade-offs: evaporative cooling systems reduce energy use but increase water demand, while air-cooled systems use less water on-site but may still rely on water-intensive energy sources.

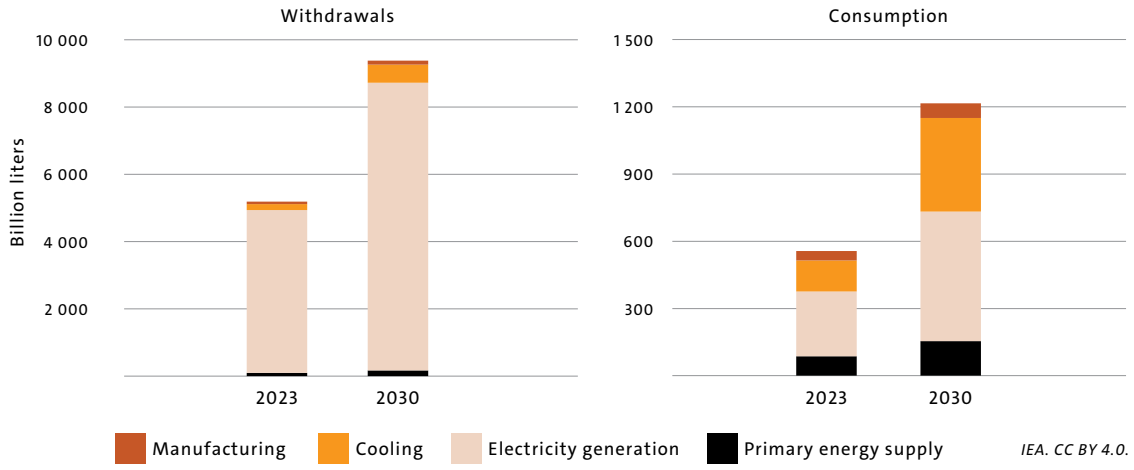
- **Direct Water Use in Data Center Cooling:** The primary source of AI-related direct water use is data center cooling. Cooling approaches vary widely, and liquid-based methods – including chilled water systems, direct-to-chip cooling, and immersion cooling – are increasingly deployed to support high-density workloads. Among these, chilled water systems remain the most widely used, particularly in hyperscale and enterprise data centers, reflecting their relative energy efficiency, technological maturity, and operational flexibility. This approach can use significant amounts of water. Techniques like evaporative cooling and air-side economizers leverage ambient conditions to lower energy use, and hybrid systems combine chilled water with dry or adiabatic cooling[9] to optimize both water and energy efficiency.

  Water Usage Effectiveness (WUE), measured as consumption in liters per kilowatt-hour (L/kWh), is a standard metric for measuring data center water efficiency. While the average data center WUE is around 1.8 L/kWh[10], some hyperscale facilities have reached as low as 0.19 L/kWh through high-efficiency cooling strategies.[11] Since WUE varies across locations and over time, strategically choosing when and where to train large AI models can significantly reduce their water footprint. Cooling system selection is influenced by climate, facility size, computing density, regulations, and environmental and cost considerations.

- **Indirect Water Embedded in the Supply Chain:** In addition to water used for electricity generation, data centers also contribute to indirect water use through embodied water, the water required to manufacture servers and construct facilities. Semiconductor facilities, in particular, are highly water-intensive, with some consuming substantial amounts of ultra-pure water per day. For example, TSMC's north Phoenix facility is expected to consume more than 17 million gallons

of water per day, an amount that could supply roughly 57,000 households.[13] While the local environmental impact of this manufacturing can be substantial in regions hosting these facilities it represents a relatively small share of a single data center's overall water footprint. According to the IEA study[6] referenced earlier, only approximately 8% of total water consumption by data centers is attributed to semiconductor and microchip production.

***Figure 2:** Water withdrawals and consumption by data centers in 2023 and 2030*



Source: Adapted from IEA (2025).[6]

Water Impacts at Different Scales

When examining the water footprint of AI specifically, data remains limited, making it difficult to fully assess the water demands of model training and inference or their broader impacts on water resources. Nonetheless, existing studies offer some insights. For example, one study indicates that the computer power required to train a large model like GPT-3 is estimated to consume approximately 185,000 gallons (700,000 liters) of direct on-site water and 1.2 million gallons (4.7 million liters) of indirect water associated with the energy generation.[9] On the inference side, the per-query environmental impact is smaller but still measurable and – at production scale over a model's lifetime – can accumulate to substantial water amounts. Each 10-50 GPT-3 queries are estimated to directly consume around 65 milliliters of water and 435 milliliters of indirect water, for a total of 500 milliliters (0.132 gallons).[14]

To evaluate AI's water footprint further and its broader implications, let us explore more deeply the data centers' water footprint, given the availability of more data and their inextricable connection to AI operations. In the context of data centers, it is important to distinguish between water use (withdrawal) and water consumption. Water use refers to the total volume of water withdrawn or utilized by a facility. Not all of this water is lost, as some may be discharged back into the system. By contrast, water consumption refers to the portion of water that is permanently removed from the local system and is no longer available for reuse – primarily through evaporation during cooling processes. At the global level, data centers water footprint is growing but remains modest compared to high-consumption sectors such as agriculture, manufacturing, and energy production. The IEA[6] estimated that global water consumption for data centers for both direct and indirect water, is currently around 148 billion gallons (560 billion liters) per year, and this could rise to around 317 billion gallons (1,200 billion liters) per year in 2030. Current global water withdrawals for data centers, estimated at over 2.3 trillion gallons (over 9 trillion liters) per year, show a similar steep increase to 2030 (Figure 2).

These numbers, though significant, represent just a fraction of other industries. Agriculture dominates global water use, accounting for approximately 70% of all freshwater withdrawals, primarily for irrigation and livestock. Each year, the sector consumes an estimated 2,500 to 3,100 trillion liters of water globally.[15] Producing one kilogram of beef, for example, requires about 15,000 liters of water.[16] The textile industry is estimated to consume over 79 trillion liters of water annually.[17] The global energy sector is a significant consumer of freshwater, accounting for roughly 10% of total global freshwater withdrawals. In 2021, the world's energy system used approximately 370 trillion liters of freshwater.[18]

While the global water use of data centers remains relatively modest in comparison to other sectors, the more pressing concern lies in their local impacts. The rapid and uncertain expansion of these facilities risks straining community water systems, particularly in arid or water-stressed regions where pressures on limited supplies are already acute. In the United States, where more than 5,400 data centers operate, of which about 20% are expected to be drawing water from moderately to highly stressed watersheds in the western U.S.[19] estimates suggest they consume roughly 0.3% of the total public water supply for the contiguous U.S.[20] Yet, these facilities can put severe pressure on water systems in the local communities that house them. For example, in The Dalles, Oregon, a Google data center in 2021 withdrew over 355 million gallons, which represented about 29% of the city's total water demand that year.[21] Many of these facilities rely on potable municipal water for cooling, competing directly with households and other industries and adding additional pressure to the local water systems. To meet the increased demand and future water needs, Google committed $28.5 million to upgrade The Dalles' water infrastructure. Large operators (e.g., Amazon/AWS, Google, Microsoft, Meta), in particular, are starting to invest more in local water systems as a way to ensure reliable access to water to support their operations and minimize impact on local communities. In addition, they are increasingly shifting toward reclaimed or non-potable sources – such as treated wastewater or regional reuse systems – to reduce dependence on municipal supplies.

A key challenge in managing data center water use is their reliance on aging municipal water systems not built for high-intensity industrial demand. Fragmented governance, regional disparities in water rights, and outdated infrastructure further complicate planning and the sustainable integration of data center water demands into the existing infrastructure and

systems. The U.S. water system alone includes nearly one million miles of pipelines, many of which are deteriorating, managed by over 50,000 utilities.[22] U.S. water systems lose an estimated 2.46 trillion gallons (9.3 trillion liters) of treated water each year due to leaks and breaks – costing utilities roughly US $6.4 billion.[23] This loss is far greater than the 135.5 billion gallons (512.6billion liters) consumed by U.S. data centers in 2018.[24] Introducing high-demand users like data centers without proper planning risks placing additional strain on already fragile infrastructure. At the same time, AI-driven leak detection and repair present a major opportunity, as reducing inefficiencies could conserve more water than data centers currently consume, potentially offsetting their added demand.

Reducing AI's direct water footprint will partly depend on transitioning data centers away from water-based cooling systems to technologies that use minimal or no water, such as air cooling or liquid immersion cooling. This will reduce its local water footprint. Additionally, transitioning away from water-intensive energy sources is critical for addressing the indirect water consumption that data centers drive. As the adoption of renewable energy continues to expand and supplies a growing share of data center operations, the overall indirect water footprint of AI is expected to decline. Since many conventional energy sources, such as coal and nuclear power, are highly water-intensive, shifting toward solar, wind, and other low-water alternatives can significantly reduce indirect water use. However, because data centers require uninterrupted, high-reliability power, a full transition to renewables also depends on advances in energy storage and grid stability. To enable continuous operation without relying on water-intensive backup sources, innovation in battery storage, smart grid systems, and hybrid renewable infrastructure will be essential.

Transitioning away from water-intensive energy sources is critical for addressing the indirect water consumption that data centers drive.

The Need for More Water Use Data

As stated, public data on the water footprint of AI infrastructure is limited and fragmented. The IEA notes that water consumption in data centers – especially for cooling and power generation – is poorly monitored and underreported.[6] A 2021 Uptime Institute survey found that only 51% of data center operators track water use, mostly at individual sites, and just 10% do so across all data center facilities.[24] The same report estimates that over 60% of companies see no business case for collecting detailed water-use data. When they do, they rarely disclose facility-level information, instead reporting aggregated corporate figures making it difficult to assess the specific water intensity of AI-related operations. While some companies have begun to increase transparency – for example, AWS, Google, and Microsoft have pledged to publish Water Usage Effectiveness (WUE) data – progress remains uneven. The lack of widely adopted, granular standards compounds the challenge. Without consistent metrics or regulatory requirements it is difficult to compare water footprints across companies or AI workloads

Pressure to disclose and reduce water use is growing, however, as an increasing number of municipalities are requiring new data centers to limit or eliminate direct water consumption. States in water-stressed regions like Arizona, Texas, and California have already paused or restricted new data center projects pending sustainable water strategies.[25] These local actions may signal the beginning of a broader regulatory shift, as AI workloads expand and the need to safeguard local water resources becomes more urgent. At the same time, investors and credit rating agencies are increasingly incorporating water-related risks into ESG frameworks, adding further pressure on data center operators to reduce their reliance on water, especially in water-stressed regions. Major agencies like Moody's and S&P Global have begun factoring water stress into credit assessments, particularly in areas where infrastructure demands and competition for limited water supplies are intensifying.[26, 27]

More broadly, the central challenge is uncertainty, which makes it hard for local and regional agencies to plan for surging water demand. Utilities typically project water needs 20-30 years ahead, and many big-ticket assets are designed for 30-50-year service. By contrast, data-center planning timelines range from a few months for retrofits to about 5-10 years for hyperscalers, creating a mismatch in planning horizons that complicates sizing, permitting, and financing of water infrastructure. Critical unknowns include how many more data centers will be built, where they will be located, and how much water they will consume both at average and peak capacity. Added to these are uncertainties about how AI technologies will evolve, the pace of innovation in reducing data center water use, and potential regulatory shifts away from water-intensive energy sources – all of which further complicate planning. These uncertainties hinder the ability to evaluate and manage long-term impacts on local water systems in communities hosting these facilities, limiting the development of effective resilience strategies. This underscores the urgent need to align data center growth with local and regional water resource planning, environmental policy, and infrastructure resilience. Achieving this alignment depends on greater transparency and standardized reporting to enable accurate assessment and proactive management of data centers' water impacts.

Strategies for AI's Sustainable Water Use

Minimizing AI's water footprint is challenged by a range of system-level barriers. Most are associated with data center current operational practices, including a heavy reliance on water-intensive cooling and energy sources, dependencies on legacy water and energy systems that are not designed for the additional demands, and limited reporting on water use. Regulatory frameworks often focus on energy efficiency rather than water impacts, while market incentives rarely penalize high water consumption. Additionally, many AI models are not optimized for efficiency, and data centers are frequently located in water-stressed regions due to factors like tax incentives or energy availability. Together, these barriers make it difficult to align AI development with sustainable water management.

To address these challenges, a multi-pronged strategy is essential – one that advances technological innovation, strengthens water resource management practices, and improves transparency through standardized disclosure of water-use metrics. Such metrics are vital for effective benchmarking, reporting, and long-term planning. Overcoming systemic barriers will also require robust collaboration between the public and private sectors to better align regulatory frameworks, investment priorities, and water governance. Ultimately, these efforts must be coordinated across the entire ecosystem of stakeholders – including technology providers, regulators, and local communities – to ensure that AI development aligns with sustainable water stewardship.

1. Foster Technological Innovation:

- **Promote Cooling Innovation and Efficiency:** To support sustainable AI deployment, innovations in cooling are essential, and significant progress is already underway. Liquid and direct-to-chip cooling offers major energy and space savings over traditional air systems. Though only 22% of organizations

currently use direct liquid cooling,[28] this process can cut cooling energy use by 30-50%.[29] For example, Microsoft launched a new datacenter design that, by adopting chip-level cooling solutions, delivers precise temperature control without water evaporation, avoiding the need for more than 125 million liters of water per year per datacenter.[30] Immersion cooling, which submerges servers in conductive liquids, has been shown to achieve up to 96% cooling efficiency.[31] Climate-appropriate systems like adiabatic cooling and air-side economizers can reduce energy use by up to 60%.[32] Hybrid systems that switch cooling modes improve water and energy efficiency. There is also strong evidence that raising server inlet (room) temperatures – within ASHRAE-recommended limits – reduces data-center cooling energy and, where evaporative systems are used, lowers cooling-water consumption. A widely cited rule of thumb is a ~4-5% reduction in energy use for each 1 °F increase in server inlet temperature.[33] In addition, AI-driven optimization is playing a growing role in data center efficiency.

- **Optimize Hardware and Software:** Data center optimization goes beyond cooling, with major sustainability gains from advances in AI hardware and software. Techniques like model compression and pruning can reduce energy use by 3x to 7x also lowering cooling-related water demand[34] Software efficiency has been a big focus of the Green AI movement which promotes models that balance accuracy with energy efficiency. On the hardware side, custom AI chips like Microsoft's 2024 models significantly improve efficiency. The Data Processing Unit (DPU) offers 4× performance improvements while consuming 3× less power than Microsoft's prior hardware.[35]

- **Set-Up Benchmarking Exercises:** Implementing robust benchmarking – systematically measuring and comparing direct and indirect water use across AI models, hardware, software, and data center environments – is essential for promoting best practices and advancing water efficiency across diverse facilities and workloads. Critically, reporting must distinguish water withdrawals from consumptive use (depletions) and quantify the share effectively removed from the local basin, so impacts are assessed accurately – not just volumes moved through the system. These efforts should also differentiate between operational phases, such as training versus inference, and account for variations in data center types, including hyperscale, edge, and co-located facilities, as well as geographic and climatic conditions. Benchmarking not only facilitates cross-industry comparisons but also encourages transparency and accountability. Integrating real-time feedback mechanisms into operations allows facility managers and AI developers to track performance trends, identify inefficiencies, and deploy targeted water-saving strategies such as AI-optimized cooling systems and workload scheduling that minimizes peak water demand. This continuous cycle of measurement, analysis, and refinement fosters a culture of innovation.



Electricity generation is one of the largest water-using sectors.

2. Strengthen Water Resources Management

- **Align Data Center Expansion with Local and Regional Planning:** Closer coordination between data center operations and local and regional water management is essential. A recent U.S. National Academies workshop report[36] warns that data centers are expanding into water-stressed regions without sufficient planning, heightening environmental risks. To address this, data center's sustainability goals must be aligned with local and regional planning through a systems-of-systems approach that views water as an interconnected resource across sectors and scales. Requiring Environmental Impact Statements (EIS) for new data centers – detailing projected water withdrawals, consumption (e.g., evaporative loss), discharge, and energy demand over the facility's lifespan – would support more transparent, sustainable development. EU's Environmental Impact Assessment (EIA) Directive requires environmental assessments for large industrial projects that may significantly affect the environment, which may include data centers, especially hyperscale facilities with high water and energy demands. However, in the United States, EIS is only triggered if a federal agency is involved (e.g., federal land, federal funding, or federal permits). Most private data centers on private land are exempt from this requirement. Some states (e.g., California, New York) have environmental assessment requirements, but most states have weak or no requirements before such projects are approved. Broader adoption of EIS or similar assessments can better support integrated infrastructure planning and help mitigate AI's growing water footprint.

- **Adopt Circular Water Economy Principles:** Adopting a circular water economy is essential to cutting freshwater use in data centers. Key strategies include closed-loop cooling, wastewater reuse, and rainwater harvesting.[37] Microsoft uses "zero-water evaporation" systems in Phoenix and Wisconsin, while Google's Georgia site recycles treated wastewater onsite. In Quincy, Washington, a Microsoft data center recirculates treated water reducing reliance on local potable groundwater. In Ireland, hyperscale data centers operated by AWS and Microsoft employ rainwater-harvesting systems to offset cooling-water and other non-potable demands. This on-site source reduces municipal withdrawals and mitigates stress on local resources.

3. Improve Transparency Through Standardized Disclosure

- **Standardize Water Use Tracking Metrics:** Effectively minimizing AI's water footprint requires development, consistent tracking, and transparent disclosure of standardized Key Performance Indicators (KPIs) that account for both data center operations and AI-specific workloads. Equally important is to provide a clear distinction between direct water use (e.g., for cooling, with immediate local impacts) and indirect water use (e.g., through electricity generation or supply chains), since each carries different implications for management and mitigation. For direct water use, key metrics such as Water Usage Effectiveness (WUE), Total Direct Water Use (the total volume of water consumed on-site), and Cooling System Efficiency (e.g., liters per ton-hour of cooling) provide a facility-level baseline. Equally important is tracking cooling water losses – including evaporation, drift, and blowdown – as well as documenting the source of water used (potable, reclaimed, or freshwater). Together, these indicators are critical for assessing both operational efficiency and the broader sustainability impacts of data centers. Useful indirect (off-site) water use/consumption metrics may include Electricity-Water Intensity of Power Source (water consumption per MWh for the electricity mix powering the data center), and Embodied Water in Hardware (water used in manufacturing and transporting servers, chips, cooling infrastructure). Although these are helpful

at the facility level, they generally lack model-level granularity, required to understand the actual impact on AI on water resources. Breaking those down into more precise metrics – like liters of water used per AI training run, inference task, consumption per GPU-hour, or liters per teraflop – can enable meaningful comparisons across AI models and platforms. Existing standards like ISO 14046:2014 and emerging ones like the IEEE Standards Association may provide initial guidance. To enable a higher level of granularity, data centers must integrate Building Management Systems (BMS) and Environmental Monitoring Systems (EMS) with water-specific instrumentation like smart meters, flow sensors, and Supervisory Control and Data Acquisition (SCADA) platforms.

- **Encourage Transparent Reporting:** Several data center operators have started to report their water use – primarily in aggregate and sometimes at a regional/site level. Standardized reporting frameworks are essential to managing AI's water footprint. Reporting should, when possible, differentiate between water withdrawals and water consumption to better help understand the true impact to the local water systems. While few are AI-specific, tools like the CDP Water Questionnaire, GRI 303 Standards, and the Water Footprint Network provide strong foundations to track and disclose water risks, usage, and impacts. Organizations such as the Green Software Foundation are pushing for more holistic, lifecycle-based environmental reporting that includes water usage. Broader efforts, such as the EU's Corporate Sustainability Reporting Directive (CSRD) and the Taskforce on Nature-related Financial Disclosures (TNFD), are also pushing for water transparency in tech. To enable proactive local and regional water management and support innovation, companies should publicly disclose detailed, infrastructure- and AI model-linked data – including cooling system type, site-level water usage, local energy mix, and the embodied water of hardware – through open-access platforms, accompanied by independent verification to ensure transparency and credibility. In addition, greater supply-chain transparency is required through third-party-verified Environmental Product Declarations (EPDs) that report life-cycle GHG, energy, water footprint, critical materials, and end-of-life assumptions for major data-center products (e.g., servers, racks, cooling equipment, and microchips). Achieving this requires coordinated industry-wide effort and a sustained commitment to standardized reporting, accountability, and integration of data centers with water planning frameworks.

As AI becomes increasingly integrated into critical systems, it is essential to establish proactive environmental safeguards that address both current impacts and future risks. Navigating these uncertainties demands a multidisciplinary and forward-looking approach – one grounded in rigorous research, transparent reporting, and policy frameworks that balance sustainability with technological advancement. This responsibility should not rest solely on the shoulders of technology companies. It is a shared societal duty, involving governments, policy makers, academia, civil society, and end users, to demand and design AI systems that align with long-term environmental goals that actively protect and enhance our water resources.

Conclusion

As AI becomes more integral to environmental services, its own environmental footprint, particularly water use, remains understudied. Training and using large-scale AI models requires significant computational power, resulting in high energy consumption and substantial heat generation, often managed by water-intensive data center cooling systems that can strain local water resources. This paper examined the water footprint of AI through the lens of data centers. Although data centers' global water footprint is small relative to many other sectors, their local impacts can be substantial – especially in water-stressed basins – where new cooling demands may compete with existing users and strain already fragile systems. Equally important is the pace of growth: the rapid expansion of AI and the data-center capacity it requires is likely to intensify these pressures in the coming years. However, with intentional design, strategic planning, and cross-sector collaboration, AI can evolve not only into a more sustainable technology, but also into a powerful enabler of broader environmental stewardship. Minimizing AI's water footprint requires a multi-faceted approach, including advanced cooling technologies such as immersion cooling, modernization of aging infrastructure in partnership with utilities, and the application of AI itself for predictive monitoring and adaptive water management. In water-stressed regions in particular, adopting circular water systems – treating and reusing process water – can enhance resilience and minimize impacts. Strong governance is essential. Mandating standardized water-use metrics, public disclosure, and open-access data platforms can improve transparency, accountability, and targeted innovation. Integrating data center development and AI workloads into local and regional water management planning is critical to balance supply and demand sustainably. Embedding sustainability into AI development and deployment is not only feasible – it is necessary. As the digital economy grows alongside rising freshwater demand, AI can become part of the solution to the very environmental challenges it currently exacerbates.

To address these challenges, a multi-pronged strategy is essential – one that advances technological innovation, strengthens water resource management practices, and improves transparency through standardized disclosure of water-use metrics.

About the author

Dr. Ana Pinheiro Privette is a seasoned professional at the intersection of science, engineering, technology, and data systems, with over 25 years of experience spanning the public sector, private industry, and academia. Trained as an environmental engineer and earth scientist, she began her career in research roles at NASA's Goddard Space Flight Center and NOAA's National Centers for Environmental Information, where her work advanced the use of remote sensing for land surface temperature and hydrological modeling. Ana currently serves as the Director of Research Applications and New Research Initiatives at the University of Illinois Urbana-Champaign, where she also manages the Center for Secure Water. Her previous leadership roles include Head of Sustainability Impact Computing at Amazon Web Services and Global Lead of the Amazon Sustainability Data Initiative (ASDI). She has directed large cross-sector collaborations such as the Partnership for Resilience and Preparedness (PREP) and the White House Climate Data Initiative (CDI), coordinating efforts across numerous federal agencies and organizations. Ana is deeply committed to advancing innovation in climate resilience and sustainability by enabling data-informed decision-making.

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